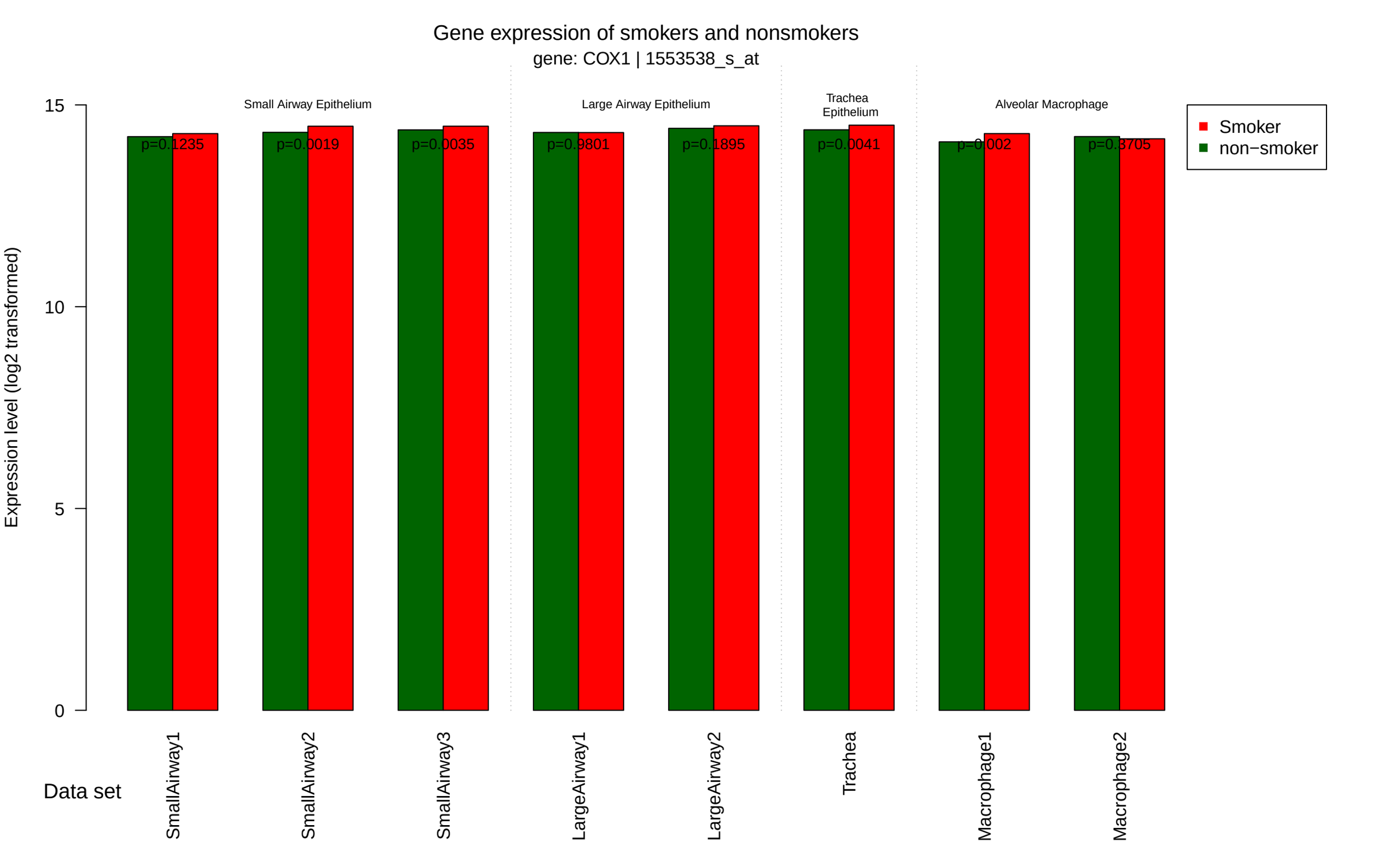
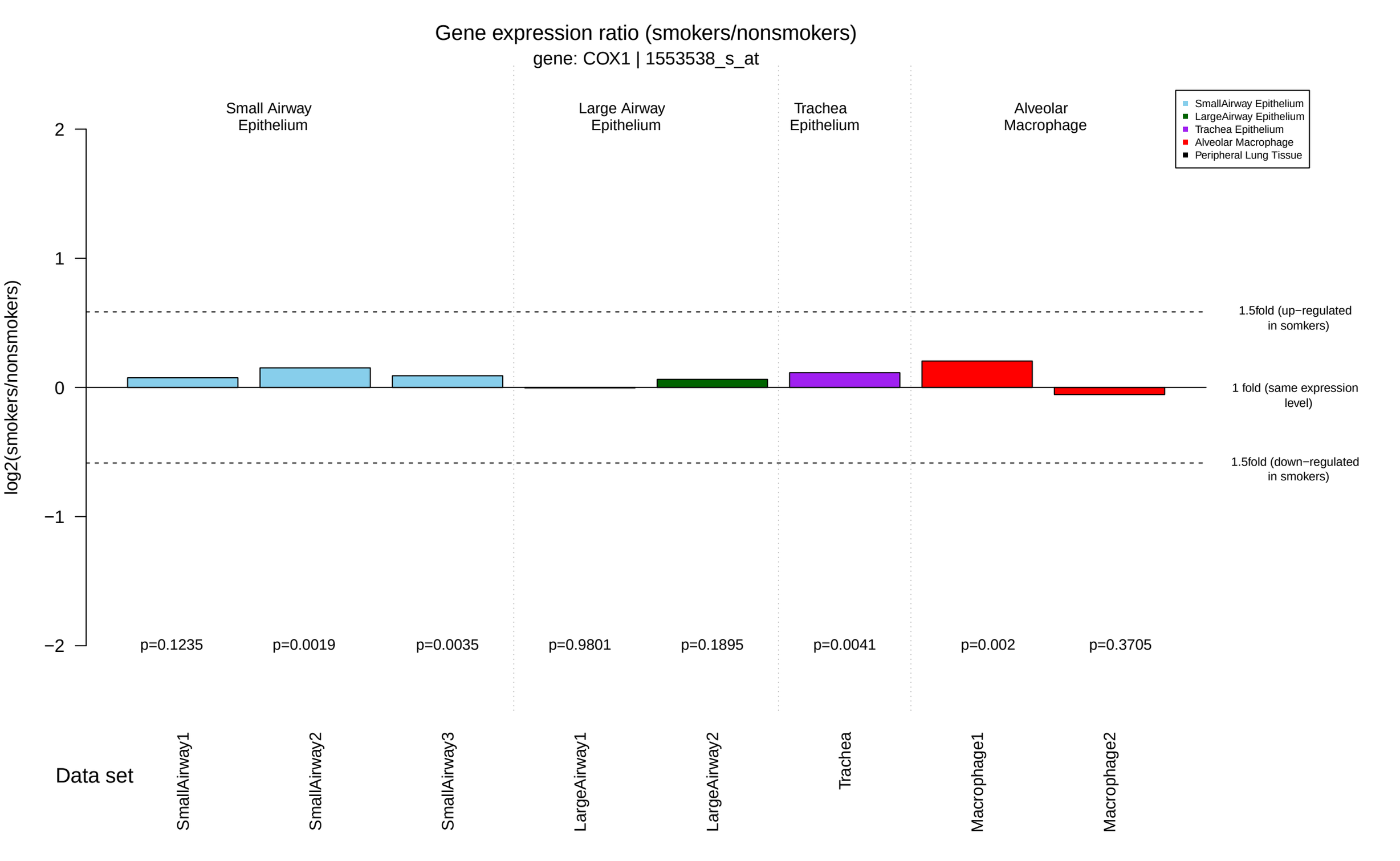
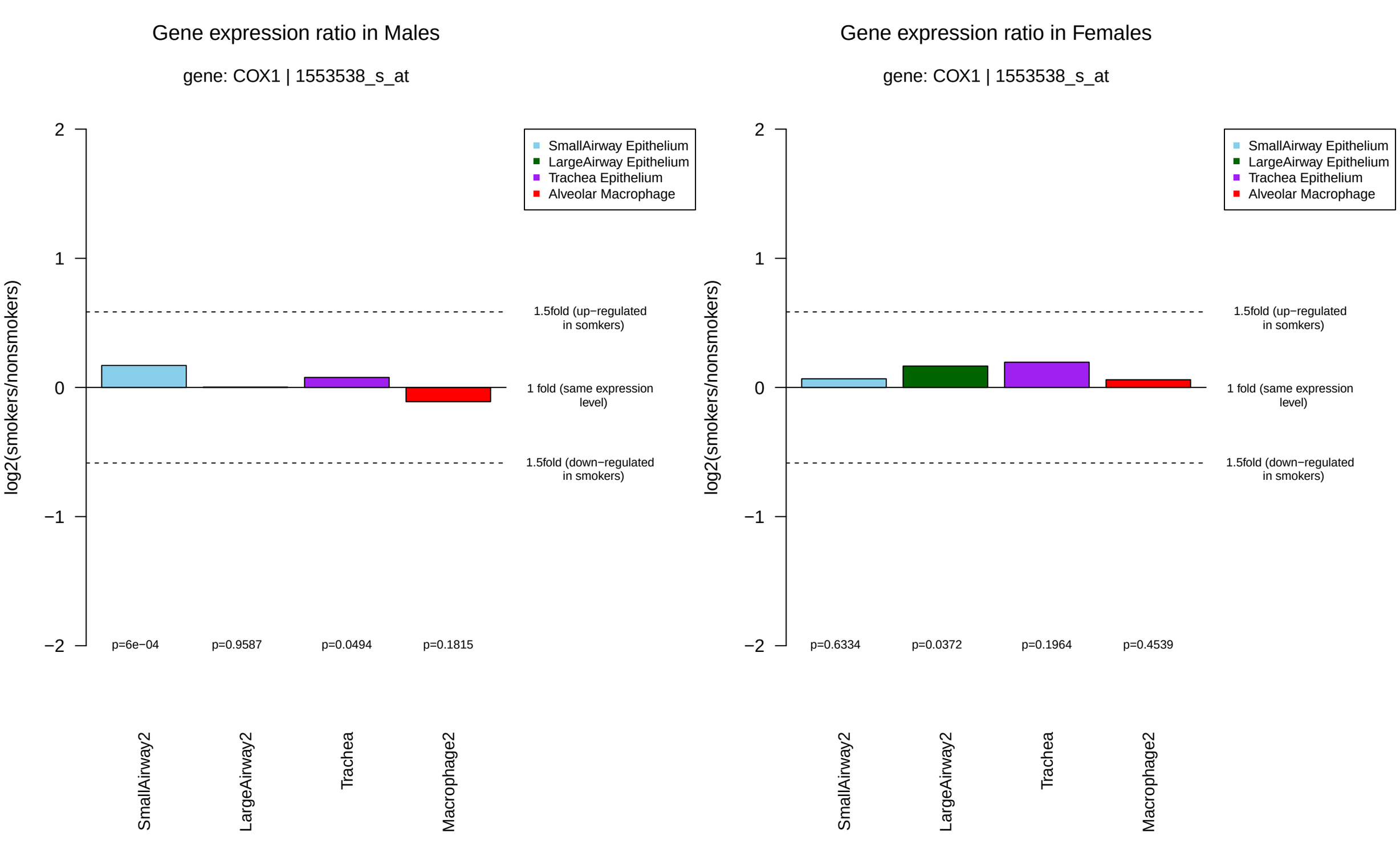
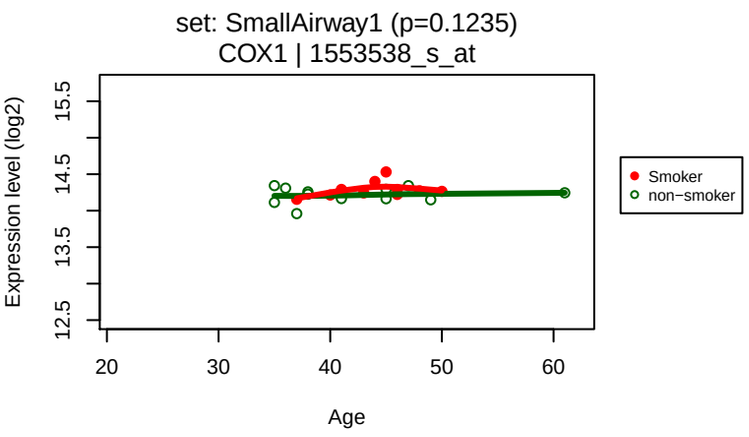
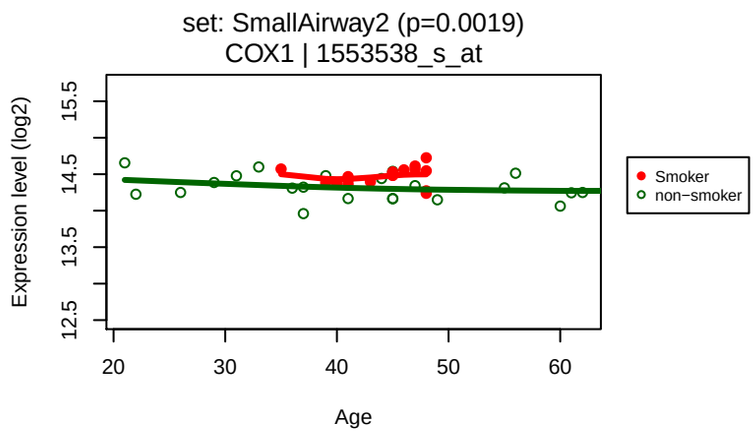
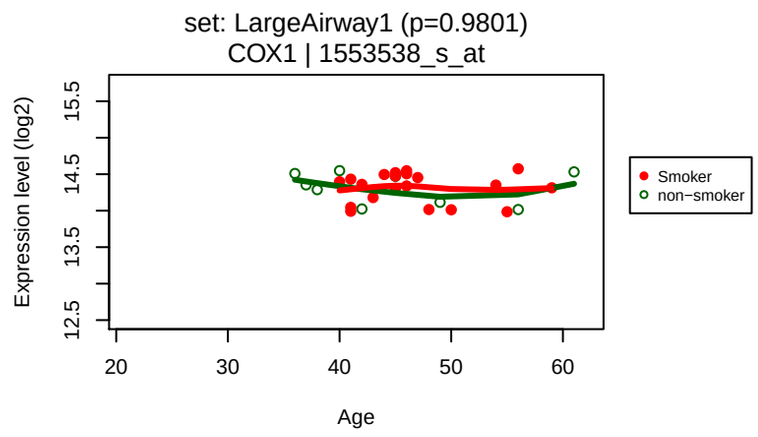

Gene expression of smokers and nonsmokers
gene: COX1 | 1553538_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1235
p=0.0019
p=0.0035
p=0.9801
p=0.1895
p=0.0041
p=0.002
p=0.3705
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

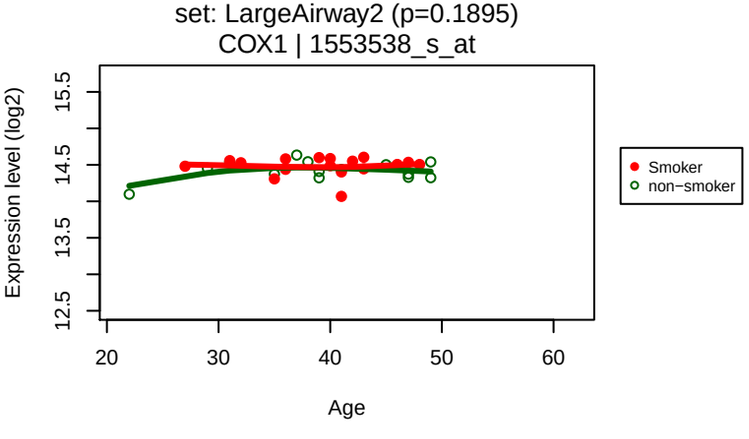

Gene expression ratio (smokers/nonsmokers)
gene: COX1 | 1553538_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1235
p=0.0019
p=0.0035
p=0.9801
p=0.1895
p=0.0041
p=0.002
p=0.3705
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

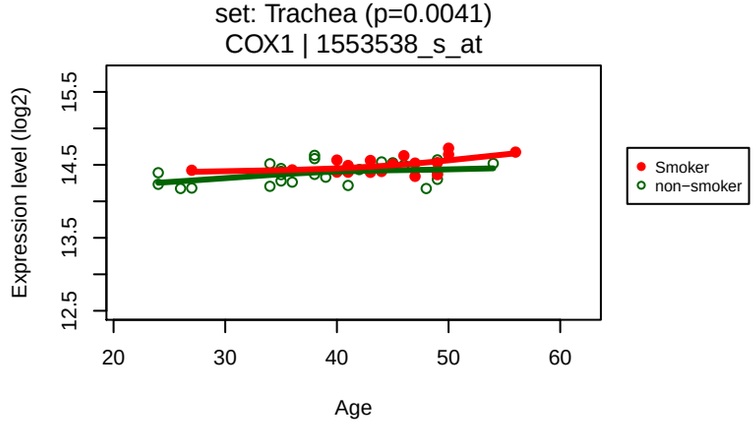

Gene expression ratio in Males
gene: COX1 | 1553538_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=6e−04
p=0.9587
p=0.0494
p=0.1815
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: COX1 | 1553538_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6334
p=0.0372
p=0.1964
p=0.4539
SmallAirway2
LargeAirway2
Trachea
Macrophage2

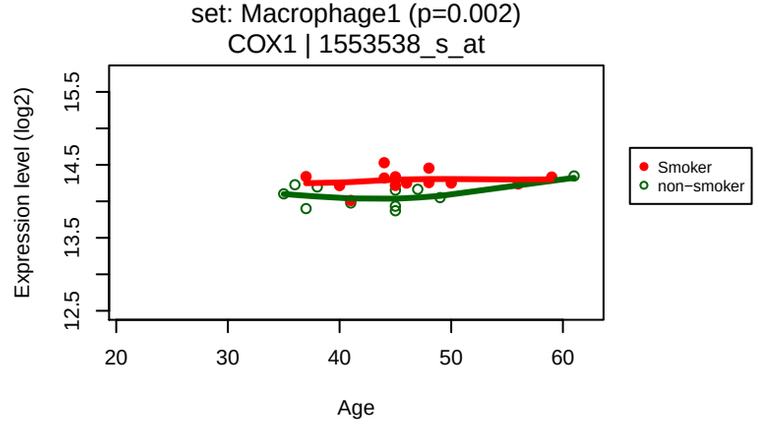

set: SmallAirway1 (p=0.1235)
COX1 | 1553538_s_at
Expression level (log2)
12.5
13.5
14.5
15.5
20
30
40
50
60
Age
Smoker
non−smoker

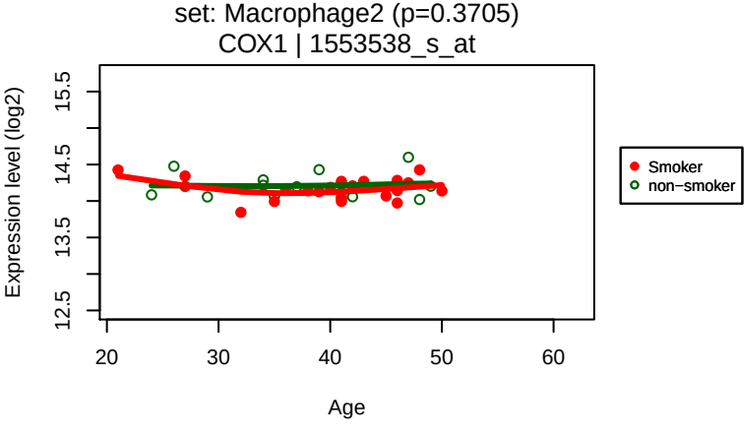

set: SmallAirway2 (p=0.0019)
COX1 | 1553538_s_at
Expression level (log2)
12.5
13.5
14.5
15.5
20
30
40
50
60
Age
Smoker
non−smoker

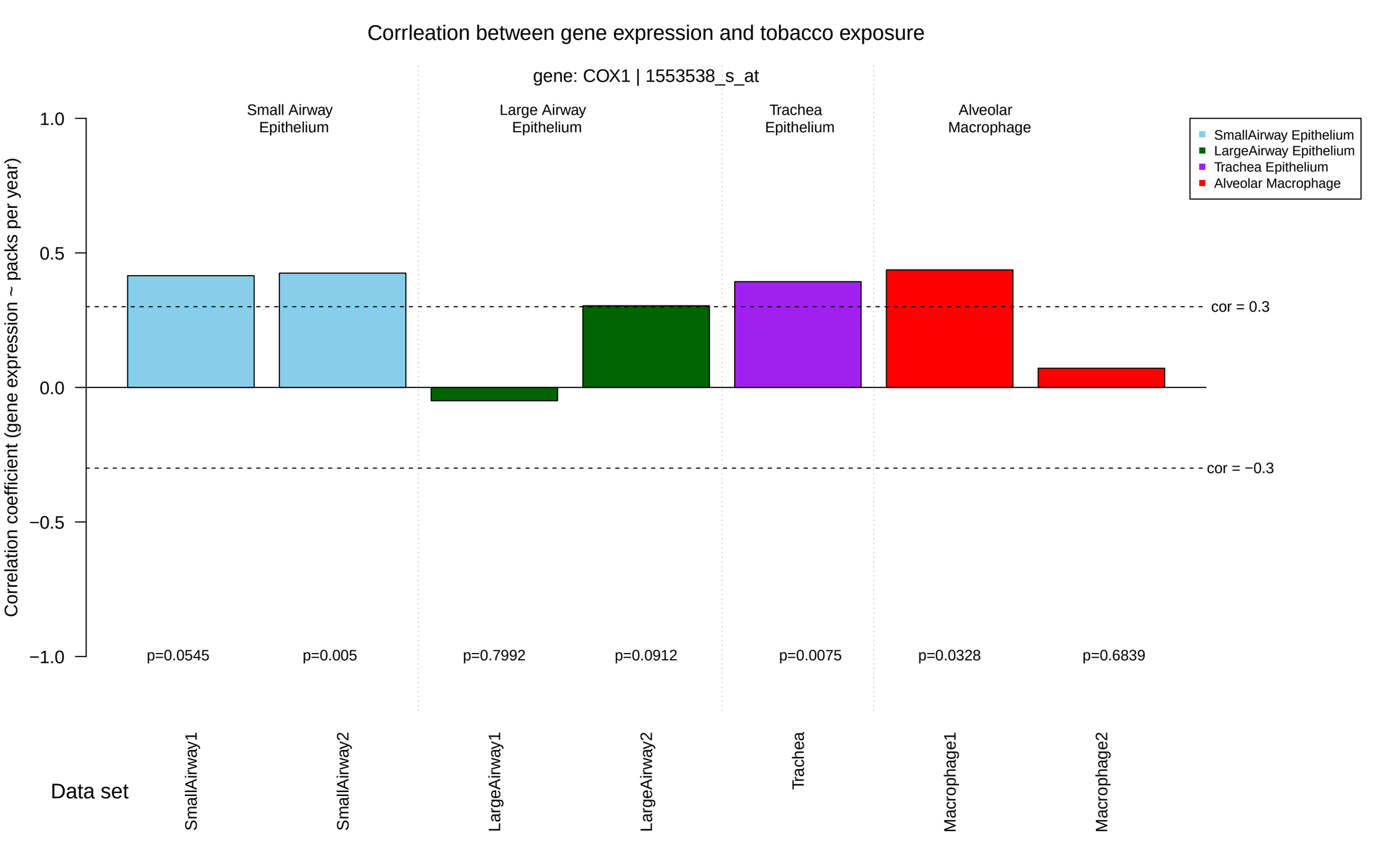

set: LargeAirway1 (p=0.9801)
COX1 | 1553538_s_at
Expression level (log2)
12.5
13.5
14.5
15.5
20
30
40
50
60
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1895)
COX1 | 1553538_s_at
Expression level (log2)
12.5
13.5
14.5
15.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Trachea (p=0.0041)
COX1 | 1553538_s_at
Expression level (log2)
12.5
13.5
14.5
15.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage1 (p=0.002)
COX1 | 1553538_s_at
Expression level (log2)
12.5
13.5
14.5
15.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3705)
COX1 | 1553538_s_at
Expression level (log2)
12.5
13.5
14.5
15.5
20
30
40
50
60
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: COX1 | 1553538_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0545
p=0.005
p=0.7992
p=0.0912
p=0.0075
p=0.0328
p=0.6839
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2